MILITARY SPECIFICATION SHEET

ELECTRON TUBE, TRAVELING WAVE

1/ TYPE 8175

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: 1-kw power output amplifier, 2.9 to 3.1 GHz, periodic-permanent magnet focused.

ABSOLUTE RATINGS: See note 1

Parameter:	Ef	eb	ik	Iw	iw	Pp	pi(rf)	Du	tp	tk
Unit:	V	kv	a	mA	ma	W	w	---	µs	sec
Maximum:	6.6	8	1.7	3.6	650	82	2	0.006	10	---
Maximum	6.0	6	---	---	---	---	---	Note 2	Note 3	120

PHYSICAL CHARACTERISTICS:

Dimensions:	See figure 1	Marking:	See note 5
DC connectors:	See figure 1	Construction:	All metal shell
RF connectors:	See figure 1	Weight:	14.5 pounds (max)
Cooling:	Forced air (Note 4)	Mounting position:	Any

TEST CONDITIONS:

Parameter:	Ef	eb	pi(rf)	Du	tp	tk
Unit:	V	kv	w	---	µs	sec
Test condition 1:	6.3	Note 6	0	0.0055	5.5	120
Test condition 2:	6.3	Note 6	0.001	0.0055	5.5	120
Test condition 3:	6.3	Note 6	0.5	0.0055	5.5	120
Test condition 4:	6.3	Note 6	1.5	0.0055	5.5	120

Frequency	
F	GHz
1	2.9
2	3.0
3	3.1

GENERAL:

(B) First article sample approval - Required (see note 7)

1/ Formerly type 334H

(B) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1	8	-					
(B) 4289	Heater current (nonoperating)	-	-	Ef = 6.3 V	If	2.2	3.2	A
(B) 1261	Electrode voltage (1) (cathode), small signal gain	6	-	F = F1 thru F3; VSWR (load) = 1.5:1 (approx)	ek	6.8	7.3	kv
(B) 1261	Electrode voltage (2) (cathode), large signal gain	6	-	F = F1 thru F3; VSWR (load) = 1.5:1 (approx)	ek	6.8	7.3	kv
(B) 1256	Electrode current (cathode)	-	3		ik	---	1.62	a
4253	Gain (1)	-	2	F = F1, F2, and F3	Gss	33	43	dB
4253	Gain (2)	-	3	F = F1, F2, and F3	Gls	33	40	dB
4253	Gain (3)	-	4	F = F1, F2, and F3	Gls	28.5	35	dB
4253	Gain variation (1)	9	2	F = F1, F2, and F3	ΔGss	---	3	dB
4253	Gain variation (2)	9	3	F = F1, F2, and F3	ΔGls	---	2	dB
4253	Gain variation (3)	9	4	F = F1, F2, and F3	ΔGls	---	2	dB
4261	Spurious output or stability	10	1		spo	---	-10	dBm
4256	Input and output match	11	-	F = F1, F2, and F3; no voltages applied	VSWR	---	2.2:1	---
4251	Output power and gain variation (1)	-	3	F = F1, eb = 1.02	ΔPo	---	1	dB
4251	Output power and gain variation (2)	-	4	F = F3; eb = 0.98	ΔPo	---	1	dB
	Quality conformance inspection, part 3							
(B) ---	Life test	-	3	Group D	t	1,000	---	hrs
---	Life-test end points	12	3	F = F2	---	---	---	---
(B) 4253	Gain (2)	-	3	F = F2	Gls	31.75	---	dB
1031	Low-frequency vibration	13, 14	-	No voltages applied	---	---	---	---
1031	Variable-frequency vibration	13, 14	-	No voltages applied	---	---	---	---
1042	Shock, specified pulse	13, 14	-	Test condition G	---	---	---	---
1011	Humidity	13, 14	-	No voltages applied	---	---	---	---
---	Shelf life	14, 15	-		---	---	---	---
(B) 1105	Permanence of marking	14	-		---	---	---	---

NOTES:

1. Cathode, one leg of heater, and focus electrode are connected internally. The anode-helix and collector are operated at ground potential.
2. The average duty cycle shall not exceed 0.006 during any 31-millisecond interval, and no pulse width shall exceed 2 percent of the pulse period.

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NOTES: -Continued

3. The pulse duration (tp) shall be defined as the time interval between the first and last instants at which the instantaneous amplitude reaches 90 percent of the peak pulse amplitude. The combination transition period of rise and fall times of the pulse, between the 10 percent and 90 percent amplitude points, shall not exceed 0.75 microsecond.

4. 25 cfm of air at sea level is required for operation at ambient temperature to 65°C.

5. The following information shall be included in the marking of each tube:

   a. Serial number.

   b. Heater voltage.

   c. Pulse beam voltage for operation with small signal and large signal in accordance with note 6. (Gss/Gls.)

   d. iwdc/iwrf' where iwrf is helix current with pi of 0.5 watt at 3 GHz, and iwdc is the helix current with pi = 0.

   e. Permanent legible inscription shall identify rf input and output connections, heater, heater-cathode, anode-helix, and ground stud.

6. The optimum pulse beam voltage for small-signal gain and large-signal gain shall be specified for each tube and included on the identification plate of the tube. The TUT shall be operated within ±3 percent of the voltage specified for the corresponding mode of operation.

7. First article sample inspection shall conform to the requirements of appendix F of MIL-E-1 and shall consist of performing all tests specified on this tube specification sheet. Invitation for bids should provide that the preparing activity reserves the right to waive the requirements for first article samples as to those bidders offering a product which has been previous procured or tested by the Government, and that bidders offering such products who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

8. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.

9. Small-signal gain variation and large-signal gain variation shall be the difference between the maximum gain and the minimum gain at the test frequencies specified.

10. The input of the TUT shall be terminated with a matched load, and the output shall be connected to a +10 dB directional coupler. The signal from the arm of the coupler shall pass through Microlab HP-FH2801 and LP-FL3201 filters, or equivalent, into a crystal detector. The output of the coupler shall be connected through a 3-dB attenuator to a variable-phase short circuit. The spurious power output shall be the maximum signal detected at the crystal as the phase of the short is varied through the one-wave length at 3 GHz.

11. The VSWR and frequency at both input and output shall be measured and recorded at the point of maximum VSWR in the frequency band.

12. The life-test end points shall be the point in time when power output falls below 750 w peak under condition of 0.5 w input anywhere within the frequency band with tube operating into a matched load.

13. During (or after) performance of specified test the TUT shall be capable of passing tests contained in quality conformance inspection, part 1.

(B)14. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

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NOTES: -Continued

15. The tubes tested and procured in accordance with this specification sheet, shall be capable of shelf-life for a period of 12 months after date of shipment from the manufacturer's plant. Tubes placed in operation during this shelf-life period, shall meet all requirements of this tube specification sheet. Tubes found to be defective, within the shelf-life period, shall be replaced by the manufacturer at no cost to the Government, or its contractors.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MI
Air Force - 80
DSA - ES

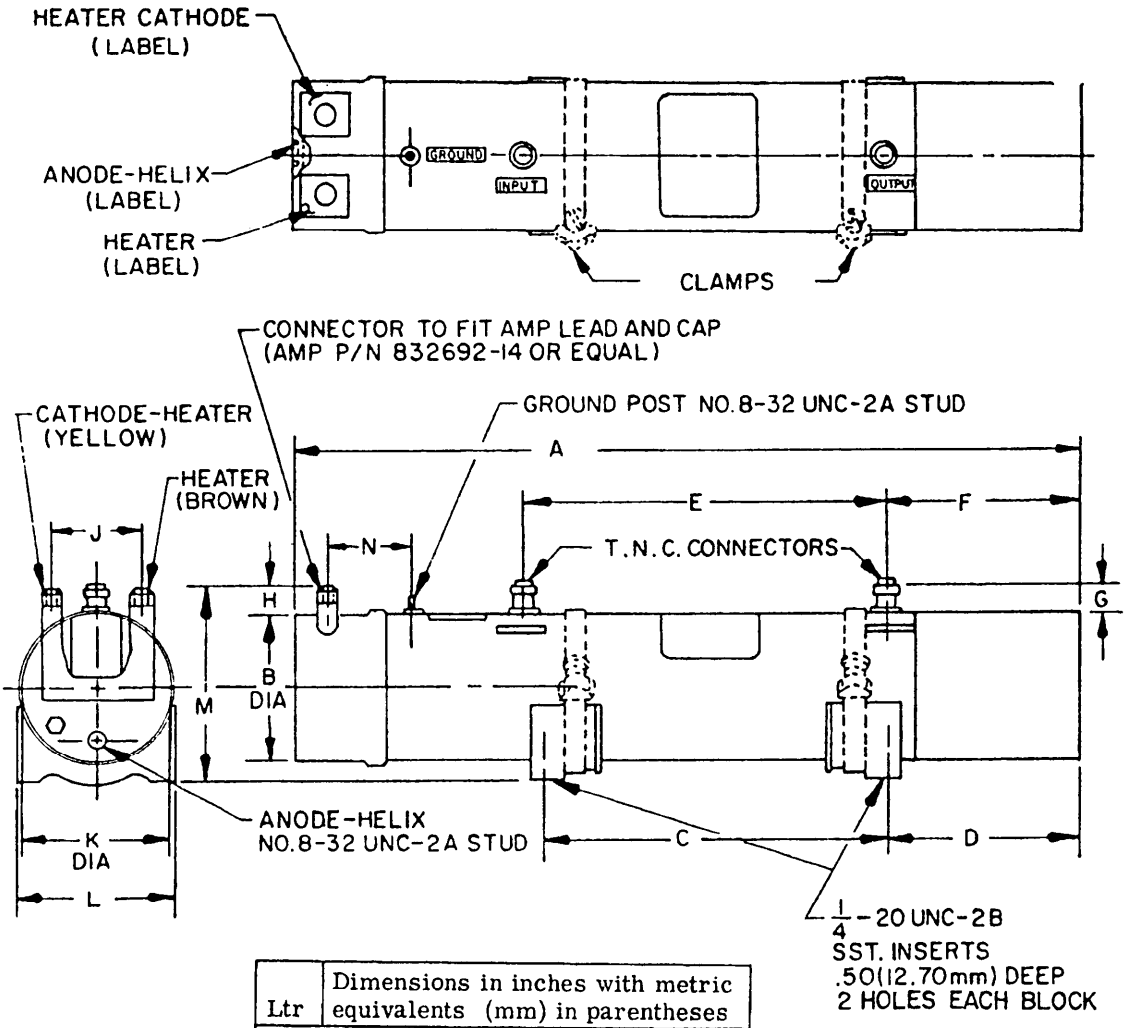
User activities:
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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
Ⓑ	Quality conformance inspection, part 1 (see note 8)	
	Minimum	Maximum
A		17.88(454.15)
B	3.59 (91.19)	3.65 (92.71)
C	7.78(197.61)	7.84(199.14)
D	4.19(106.43)	4.31(109.47)
E	7.48(189.99)	7.98(202.69)
F	4.03(102.36)	5.03(127.76)
G	.56 (14.22)	.68 (17.27)
H	.36 (9.14)	.48 (12.19)
J	1.97 (50.04)	2.03 (51.56)
K	3.36 (85.34)	3.40 (86.36)
L		3.88 (98.55)
M		4.82(122.43)
N	1.63 (41.40)	1.87 (47.50)

FIGURE 1. Outline drawing of electron tube type 8175.

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